

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

OPTICAL FIBER COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1. (a) A multimode step index fiber with core diameter of 80 micrometer and relative index difference is 1.5% is operating at wavelength of 0.85 micrometer. If the core refractive index is 1.48 then.
Determine :
 - (I) Numerical Aperture.
 - (II) Normalized frequency of fiber.
 - (III) Modes of fiber. 5(CO1)
 - (b) Compare Step index and graded index optical fiber.
Estimate wavelength for step index optical fiber in single mode operation. The core refractive index is 1.46 and core radius is 4.5 micrometer. The relative index difference is 0.25% 5(CO1)
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2. (a) Silica has estimated temp, 1500K with an isothermal compressibility $7 \times 10^{-11} \text{ m}^2\text{N}^{-1}$. The refractive index and photoelastic coefficient for silica are 1.48 and 0.286 respectively. Determine theoretical attenuation due to Rayleigh scattering along with loss factor in silica at optical wavelength 600 nm and 850 nm. (Boltzmann Const. = $1.381 \times 10^{-23} \text{ J/K}$). 5(CO3)

- (b) A 2X2 fiber coupler has input power 200 microwatt, throughput is 90 microwatt. The coupled power 85 microwatt and cross talk power 6.3 microwatt.
Compute
- (I) Coupling Ratio.
 - (II) Excess Loss.
 - (III) Insertion loss.
 - (IV) Cross Talk. 5(CO2)
3. (a) An Optical LED operating at 1310 nm has radiative and non radiative recombination times of 30 ns and 100 ns respectively. If 40 mA current is injected then calculate :
- (I) Bulk recombination life time.
 - (II) Internal quantum efficiency.
 - (III) Internal power. 5(CO4)
- (b) Illustrate the principle of Fabry-Perot LASER and Distributed feedback LASER. 5(CO4)
4. (a) An Avalanche Photo Detector (APD) has quantum efficiency at wavelength of 900 nm. If 0.5 microwatt of optical power produces multiplied photocurrent of 10 microA, compute
- (I) Responsivity of APD.
 - (II) Unmultiplied photocurrent.
 - (III) Multiplication factor. 5(CO4)
- (b) Compare PIN and APD photo detectors used in optical fiber communication. Explain with neat sketch, front and side illuminated PIN photodiode. 5(CO4)

5. (a) When mean optical power launched into an 8 KM length of fiber is 12 microwatt, then mean optical power at fiber output is 0.3 microwatt. Compute
- (I) Overall Signal attenuation.
 - (II) The overall signal attenuation for 10 KM optical link using same fiber splices at 1 KM each providing attenuation of 1 dB.
- 5(CO3)
- (b) Enlist various factors for Scattering losses.
Explain in brief Linear and non linear Scattering losses. 5(CO3)
6. Explain any **Two** in detail :
- (I) Wavelength Division Multiplexing (WDM) in optical fiber communication.
 - (II) SONET / SDH Rings and Networks.
 - (III) Link Power Budget. 10(CO5)

